DS 501 · INTRODUCTION TO DATA SCIENCE

Week 02

Data Science Methodology

How a data science project moves from a business problem to a deployed, monitored model, and how CRISP-DM organises the work.

THIS WEEK

2-1	**From Problem to Data Collection**	45 minutes
2-2	**From Understanding to Feedback**	45 minutes
2-3	**CRISP-DM and Other Process Models**	45 minutes
P2	**Practice: CRISP-DM walkthrough on telecom churn**	60 minutes · Python notebook

International Balkan University · [Instructor name] · [Semester]

Recap of Week 1

Three ideas from week 1 are the foundation of this week's methodology:

Start with a question

Data science starts with a **decision** someone must make, not with a dataset or a tool. An actionable question names a decision, a target and a time frame.

The process is a loop

Clarify the question, identify the data, analyse and model, communicate. Results raise new questions, so projects go round the loop more than once.

Explain or predict

Breiman's two cultures: some questions need a model that **explains**, others a model that **predicts** new cases accurately.

This week each of these ideas becomes a named stage: business understanding, iteration and feedback, and the analytic approach.

DS 501 · WEEK 2 · LESSON

2-1

From Problem to Data Collection

Lesson 1 of 3

In this lesson

01 Why data science projects need a methodology

02 The ten stages of IBM's Foundational Methodology

03 Business understanding and the analytic approach

04 Data requirements and data collection

45 minutes

Builds on lesson 1-2: the process

Objectives

By the end of this lesson, you will be able to:

01 **Explain** why data science projects need a methodology

Checked in: Quiz 2 · question 1

02 **Describe** the ten stages of IBM's Foundational Methodology

Checked in: Quiz 2 · question 2

03 **Apply** business understanding to a new problem

Checked in: Quiz 2 · question 3

04 **Decide** which analytic approach fits a question

Checked in: Quiz 2 · question 4

05 **Identify** data requirements and plan their collection

Checked in: Quiz 2 · question 5 and Practice 2

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Case: “The Model Nobody Used”

THE SITUATION

A retail bank's data team spent four months building a model that predicts which customers will close their accounts. On the test data it was right in most cases.

> “Six months later, nobody in the branches uses it. The model is good. What went wrong?”

The team leader asks you to find the cause before the next project starts.

YOUR TASK · PAIRS · 5 MIN

1. List three reasons, other than the model itself, why a good model may not be used.

2. For each reason, say when in the project it should have been noticed.

3. What would you ask the branch managers on the first day of the next project?

What Is a Data Science Methodology?

- A **methodology** is a general strategy that guides the stages and activities of a project, independent of any tool or technology.
- It divides a project into stages, each with a clear question to answer and an output to produce.
- It defines checkpoints where the team can stop, change direction or go back.
- It gives the sponsor, analysts and engineers a shared vocabulary for “where are we?”

Two well-known methodologies

- IBM Foundational Methodology for Data Science (Rollins, 2015): ten stages
- CRISP-DM (2000): six phases, the most widely used
- Same core ideas: question first, iterate, evaluate against the business

Key point: a methodology is the **same** whether the team uses Excel, Python or a cloud platform.

Objective 01

Why Data Science Projects Fail

Each common failure has a stage of the methodology that is designed to prevent it.

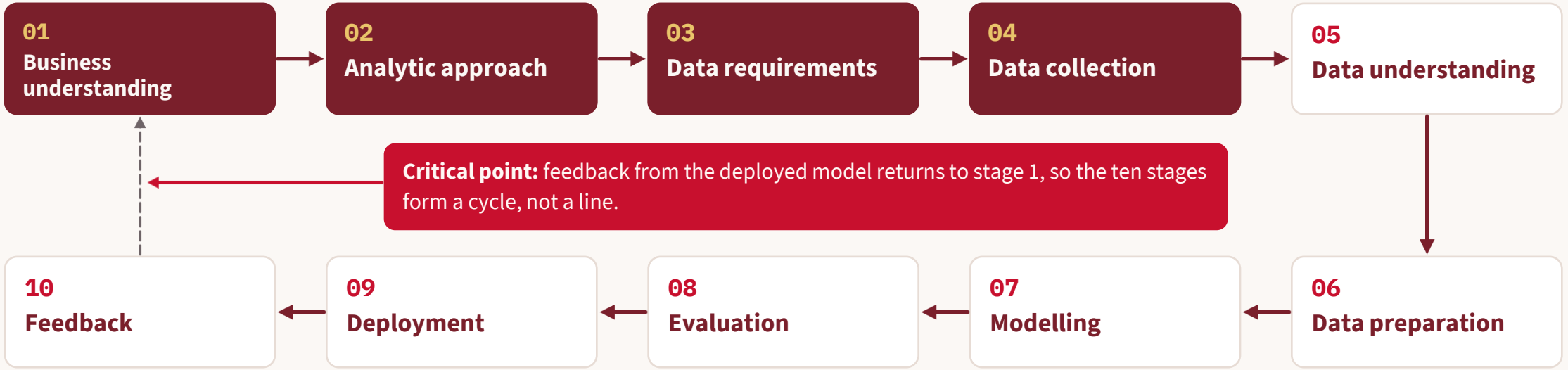
Failure	What it looks like	Stage that prevents it
Wrong question	An excellent model answers something nobody needs	1 Business understanding
Wrong data	The data that happens to be available decides the analysis	3 Data requirements
Misleading result	High accuracy, but no better than a simple guess	8 Evaluation
No path to use	Results stay in a notebook; nobody owns the roll-out	9 Deployment
No learning loop	The model decays as customers or rules change	10 Feedback

In practice

Map each row to a project you know. Most real failures are in rows 1, 4 and 5.

Read the first row first. A wrong question cannot be repaired by better data or a better algorithm.

IBM Foundational Methodology: Ten Stages



- Stages **1–4** (filled) are this lesson: from the problem to the collected data.
- Stages **5–10** are lesson 2-2: from understanding the data to feedback.

Running Case: Heart-Failure Readmissions

The IBM course Data Science Methodology follows one case through all ten stages:

Problem

Patients with congestive heart failure are often readmitted soon after discharge, which is costly for the organisation and harmful for patients.

Question

How can a limited care budget be spent so that it prevents as many readmissions as possible?

Approach

Predict each patient's risk of readmission with a model clinicians can read: a **decision tree**.

Use

At discharge, the risk score decides who receives follow-up care; outcomes are fed back to improve the model.

Source: IBM, Data Science Methodology (Coursera). The case is retold in our own words.

Objective 03

Stage 1: Business Understanding

- Every project starts by understanding the **business problem**; this stage sets the direction for all the others.
- Break the **goal** into objectives that data can support, and agree how success will be measured.
- Identify the **sponsor** and the users, and keep them involved throughout the project.
- Keep asking questions: sponsors often describe a symptom, not the underlying problem.

Outputs of stage 1

- Goal and objectives
- Sponsor, stakeholders, users
- Success criteria
- Constraints: budget, time, law

⚠ **Common mistake:** treating business understanding as one meeting. It is revisited whenever evaluation or feedback shows the question has changed.

Objective 03

Business Understanding Checklist

Use these six items in every project; the right-hand column applies them to the running case.

Item	Question to ask	Heart-failure example
Goal	What does the organisation want to change?	Fewer readmissions of heart-failure patients
Objective	What must we know or predict to get there?	Which patients are at high risk at discharge?
Sponsor	Who decides and who pays?	Director of clinical services
Users	Who acts on the result, and how?	Discharge nurses arrange follow-up care
Success	Which number should change, and by how much?	Lower 30-day readmission rate [target agreed]
Constraints	What limits our options?	Health-data law; explainable model; nurse time

In practice

This checklist is the first page of your week 7 project proposal.

Critical point: agree the success criterion **before** modelling, or any result can be called a success.

Objective 03

A model is only as useful as the decision it changes.

Accuracy, speed and elegance matter only if someone acts differently because of the result. Business understanding names that decision and that person before any data is collected.

- Rule of thumb: if you cannot name who will act on the result and what they will do, go back to stage 1.

Objective 04

Stage 2: Choosing an Analytic Approach

The **analytic approach** expresses the business question as a type of analysis. The type of question decides it.

Type of question	Analytic approach	Example
What happened?	Descriptive statistics, charts	Readmissions per month last year
Why did it happen?	Diagnostic: statistical tests	Is readmission linked to age?
Will it happen? (yes / no)	Classification	Will this patient be readmitted?
How much or how many?	Regression or forecasting	Beds needed next week
Which cases are similar?	Clustering	Groups of patients with similar care needs
What should we do?	Optimisation, decision analysis	Whom to follow up with 20 nurse visits

Why? The case needs a yes/no prediction that clinicians can read, so it chose a classification tree.

⚠ **Common mistake:** choosing a fashionable technique first (“let's use deep learning”) and fitting the question to it afterwards.

Exercise: Which Analytic Approach?

Choose the most suitable approach for each question. Work in pairs for 4 minutes.

Question	Descr.	Classif.	Regr.	Cluster.
How many students enrolled in each IBU faculty last year?				
Will this loan applicant miss a payment in the next 12 months?				
How many kWh will Skopje use next Tuesday?				
Which groups of shoppers behave in similar ways?				
Is this incoming e-mail spam?				
What was the average delivery time by region in March?				

Why? First ask: past or future? Then: a category, a quantity, or groups?

Tip: a question about an unseen case counts as prediction, even when it is not about the future.

Stage 3: Data Requirements

- The analytic approach decides which data are needed, as a recipe decides the **ingredients**.
- Define the **unit of analysis**: what one row represents, such as one patient or one customer.
- Set **inclusion and exclusion criteria** and the time window the data must cover.
- Define the **target variable** precisely, and list the variables that could explain it.

Case: data requirements

Unit: one patient with a heart-failure admission in the study period.

Exclude: patients whose readmission risk is driven by other conditions [criteria agreed with clinicians].

Target: readmitted within 30 days of discharge, yes or no.

Inputs: demographics, diagnoses, medication, prior admissions.

⚠ **Common mistake:** writing “all the data” as the requirement. It names no unit, no target and no time window, so gaps stay hidden.

Stage 4: Data Collection for the Case

Collection gathers each requirement from its source and records what is missing.

Requirement	Example variables	Likely source
Demographics	Age, sex, area of residence	Patient register
Clinical history	Diagnoses, other conditions, lab results	Electronic medical record
Medication	Drugs prescribed and dosage	Pharmacy system (partly missing)
Prior use	Admissions in the past year, length of stay	Admissions database
Target	Readmitted within 30 days: yes / no	Derived from admissions

⚠ **Common mistake:** collecting personal health data before checking the legal basis for using it.

← **Critical point:** a missing source is a decision: proceed and add it later, or stop.

Objective 04

Check Your Understanding

A hospital wants to know, for each patient discharged today, whether the patient will be readmitted within 30 days, so that nurses can follow up the riskiest patients. Which analytic approach fits best?

A Descriptive statistics of last year's readmission rates by ward.

B Classification: predict readmitted yes or no for each new patient.

C Clustering: group patients with similar records, without a target.

D Regression: predict each patient's length of stay in days.

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Methodology
A tool-independent strategy of stages that guides a project.

Business understanding
Defining the goal, objectives, sponsor and success criteria.

Analytic approach
The type of analysis that the business question requires.

Unit of analysis
What one row of the final dataset represents.

Foundational Methodology
IBM's ten-stage cycle, from business understanding to feedback.

Success criterion
The measurable result that shows the project has worked.

Data requirements
The content, format and sources the approach needs.

Target variable
The precisely defined outcome a predictive model estimates.

Summary

In this lesson, you learned how to:

- **O1** Explain why data science projects need a methodology
- **O2** Describe the ten stages of IBM's Foundational Methodology
- **O3** Apply business understanding to a new problem
- **O4** Decide which analytic approach fits a question
- **O5** Identify data requirements and plan their collection

Key takeaway: the first four stages decide what the project is for and what data it needs, before any model exists.

Next lesson
2-2 From Understanding to Feedback

DS 501 · WEEK 2 · LESSON

2-2

From Understanding to Feedback

Lesson 2 of 3

In this lesson

01 Understanding and preparing the data

02 Data leakage, and why we keep a test set

03 Evaluating a model against the business

04 Deployment, monitoring and feedback

45 minutes

Builds on lesson 2-1: stages 1–4

Objectives

By the end of this lesson, you will be able to:

01 **Describe** how data understanding tests fitness for use

Checked in: Quiz 2 · question 6

02 **Explain** data preparation and data leakage

Checked in: Quiz 2 · question 7

03 **Explain** why models are tested on unseen data

Checked in: Quiz 2 · question 8

04 **Evaluate** a model against a baseline and costs

Checked in: Quiz 2 · question 9 and Practice 2

05 **Describe** deployment and what to monitor afterwards

Checked in: Quiz 2 · question 10

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Case: “Ready to Deploy on Monday?”

THE SITUATION

A data team has built a model that predicts which heart-failure patients will be readmitted within 30 days. On the test data it is right for 98% of patients.

> “98% is excellent. Can we start using it in all wards on Monday?”

You are asked to review the project before the decision.

YOUR TASK · PAIRS · 5 MIN

1. Why might 98% accuracy be a warning sign rather than good news?
2. Which two questions would you ask about the data and the variables?
3. What would you propose instead of using it in all wards on Monday?

Objective 01

Stage 5: Data Understanding

- **Data understanding** asks whether the collected data represent the problem and are of good enough quality.
- Describe every variable: summary statistics, frequency tables and histograms.
- Look at pairs of variables: correlations reveal related or redundant variables.
- Expect to loop back: what you find often changes what you must collect.

Case: a loop back to collection

The first look at the data showed that “heart-failure admission” was not defined the way the team had assumed.

The team **refined the definition** with clinicians and collected the data again.

Key point: going back from stage 5 to stage 4 is a normal part of the method, not a failure.

Six Data Quality Checks

Run these checks on every new dataset; the examples are illustrative.

Check	What to look for	Example in a hospital
Completeness	Missing values or missing records	No weight recorded for some patients
Validity	Values outside the allowed range	Discharge date before admission date
Consistency	The same thing coded differently	“M”, “Male” and “1” for one sex
Uniqueness	Duplicate records	One stay entered by two departments
Timeliness	Data too old for the question	Medication list from two years ago
Representativeness	Sample unlike the target population	One hospital, model meant for all

In practice

Week 5 turns each check into pandas code.

Critical point: the last check cannot be seen in the data alone; ask who is **not** in it.

Stage 6: Data Preparation

- **Data preparation** turns raw collected data into the table a model can use: one row per unit of analysis.
- It is often the longest stage: practitioners estimate that collection, understanding and preparation take most of a project's time.
- **Feature engineering** creates new variables that capture useful information, such as the number of prior admissions.

Main tasks

- Clean: missing, invalid, duplicate
- Standardise formats and codes
- Integrate sources on common keys
- Aggregate to the unit of analysis
- Engineer new features

⚠ **Common mistake:** preparing the data once and never documenting it. Every step must be written as code that can be run again.

Objective 02

From Hospital Stays to One Row per Patient

Admissions: one row per hospital stay

Patient	Admitted	Diagnosis	Days
P01	04 Jan	Heart failure	6
P01	19 Mar	Heart failure	4
P01	02 Apr	Pneumonia	3
P02	11 Feb	Heart failure	5
P03	27 Feb	Heart failure	2

Illustrative records; dates in one year.

aggregate →

Analysis table: one row per patient

Patient	Stays	HF stays	Mean days	Readmit 30 d
P01	3	2	4.3	yes
P02	1	1	5.0	no
P03	1	1	2.0	no

Critical point: the target and the new features are computed per patient, the unit of analysis chosen in stage 3.

- Aggregation **creates** features, such as the number of stays and the mean length of stay.
- The target, readmitted within 30 days, is also **derived** from the stays.

Data Leakage

- **Data leakage** happens when a model is trained with information that would not be available when the real prediction is made.
- Leakage makes a model look excellent in testing and fail in use.
- Prevent it: fix the **moment of prediction** and use only data known before it.
- Split the data **before** any preparation step that learns from it.

Three examples of leakage

Using “follow-up visit booked” to predict readmission, when visits are booked only for high-risk patients.

Using the final bill of a month to predict whether a customer left in that month.

Scaling with the mean of **all** rows, including the test set.

Key point: a model may use only what is known at the moment it makes its prediction.

Objective 02

If a result looks too good to be true, look for leakage first.

Real data are noisy, and real predictions are uncertain. A model that is almost always right usually knows something it should not: the answer, hidden in one of its inputs.

- Rule of thumb: for every input, ask “would I know this value at the moment of prediction?”

Stage 7: Modelling

Modelling builds the model chosen in stage 2. Models are of two broad kinds:

Descriptive models

Summarise relationships in the data that exist.

Example: patients with diabetes are readmitted more often.

Used for: understanding, explaining and reporting.

Predictive models

Estimate an unknown outcome for a new case.

Example: this patient's probability of readmission is 0.42.

Used for: acting on individual cases.

Key point: start with a simple **baseline** that every model must beat, and record each attempt.

Training and Test Sets

All 7,043 telecom customers (Practice 2)

Critical point: if the test set influences any choice, its score is no longer honest.



Fit every step that learns from data here: imputation values, scaling, category encoding and the model itself. Compare candidate models here too.

Used once, at the end, to estimate performance on new customers.

Both sets keep the same churn rate, 26.5%, because the split is stratified on the target.

⚠ **Common mistake:** splitting the data after cleaning and scaling it, so that the test set has already shaped the preparation.

Stage 8: Evaluation Beyond Accuracy

Churn model from Practice 2 on its 1,761 test customers, threshold 0.5:

	Predicted: leaves	Predicted: stays
Actually left (467)	259 caught	208 missed
Actually stayed (1,294)	133 wasted offers	1,161 correct

In practice

At threshold 0.3 the model catches 356 churners (recall 0.762) and wastes 314 offers.

Read the top row first. 208 missed churners cost far more than 133 wasted offers.

- Accuracy is 0.806 against a baseline of 0.735, but **recall** is only 0.555: the model finds about half of the customers who leave.

Key point: the best threshold depends on what each error costs; Practice 2 finds it by computing the total cost.

Stage 9: Deployment

Deployment puts the model into real use, where it informs decisions. Good practice:

01

Pilot first: one ward, one region or a subset of users, for a fixed period.

02

Integrate the output into existing work, e.g. a risk score in the discharge system.

03

Train users: explain what the score means, and what it does not mean.

04

Document the model, its data, its limits and who is responsible for it.

05

Measure the pilot against the success criteria agreed in stage 1.

In practice

Deployment involves IT, data engineers and the users, not only the data science team.

Stage 10: Feedback and Monitoring

Feedback collects results from the deployed model to measure its impact and improve it. Monitor four things:

Model performance

Recall, precision and calibration on recent cases, compared with the values at deployment.

Data drift

New cases that look different from the training data: a new hospital, a new coding system, a new tariff.

Business outcome

The number that motivated the project, such as the 30-day readmission rate, and the cost of the interventions.

Fairness and complaints

Performance by age, sex or region, and feedback from users who disagree with the model.

⚠ **Common mistake:** judging the model on post-deployment data without noticing that its own actions changed the outcomes.

Objective 02

Check Your Understanding

A churn model includes the variable “days since the customer requested cancellation”. Its accuracy on the test set is 99%. What is the most likely explanation?

A The model is overfitted because it has too many variables.

B Data leakage: the variable is known only after the decision to leave.

C The classes are imbalanced, so accuracy is always high.

D The test set is too small to give a reliable estimate.

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

Data understanding

Checking that the data are of good quality and represent the problem.

Data preparation

Turning raw data into one analysis-ready row per unit of analysis.

Feature engineering

Creating new variables that capture useful information.

Data leakage

Training with information not available at prediction time.

Baseline model

A very simple model that any useful model must beat.

Test set

Data kept apart from model building, used once to estimate performance.

Recall

The share of actual positive cases that the model finds.

Data drift

New cases differ from the data the model was trained on.

Summary

In this lesson, you learned how to:

- **O1** Describe how data understanding tests fitness for use
- **O2** Explain data preparation and data leakage
- **O3** Explain why models are tested on unseen data
- **O4** Evaluate a model against a baseline and costs
- **O5** Describe deployment and what to monitor afterwards

Key takeaway: a model is ready for use only when it beats a baseline on unseen data, without leakage, at an acceptable cost of errors.

Next lesson
2-3 CRISP-DM and Other Process Models

DS 501 · WEEK 2 · LESSON

2-3

CRISP-DM and Other Process Models

Lesson 3 of 3

In this lesson

01 Where CRISP-DM came from, and its six phases

02 How CRISP-DM maps to IBM's ten stages

03 Other process models, and criticisms of CRISP-DM

04 An ethics checkpoint for every project

45 minutes

Builds on lessons 2-1 and 2-2

Objectives

By the end of this lesson, you will be able to:

01 **Describe** the origin and six phases of CRISP-DM

Checked in: Quiz 2 · questions 11–12

02 **Distinguish** CRISP-DM from IBM's methodology

Checked in: Quiz 2 · question 13

03 **Compare** other process models with CRISP-DM

Checked in: Quiz 2 · question 14

04 **Evaluate** the main criticisms of CRISP-DM

Checked in: Quiz 2 · question 15

05 **Apply** CRISP-DM with an ethics checkpoint

Checked in: Quiz 2 · question 16 (applied) and Practice 2

Look for the objective chip in the top-right corner of each slide: it shows which goal that slide serves.

Objective 03

Case: “Which Process Should We Use?”

THE SITUATION

A bank in Skopje has hired its first four data specialists. They will build churn and credit-risk models that run every night.

> “IBM's methodology, CRISP-DM, TDSP, MLOps… Pick one way of working and tell me why.”

The bank already runs its IT projects in two-week agile sprints.

YOUR TASK · PAIRS · 4 MIN

1. What do you need to know about the team and the projects before choosing?
2. Which model, or combination, would you suggest now?
3. Revisit your answer at the end of the lesson.

Where Did CRISP-DM Come From?

- **CRISP-DM** stands for Cross-Industry Standard Process for Data Mining.
- It was started in 1996 by a consortium of companies that used data mining, and version 1.0 was published in 2000.
- It was designed to be **industry-neutral**, **tool-neutral** and based on real project experience.
- More than 25 years later, it is still the most widely cited process model.

The founding consortium

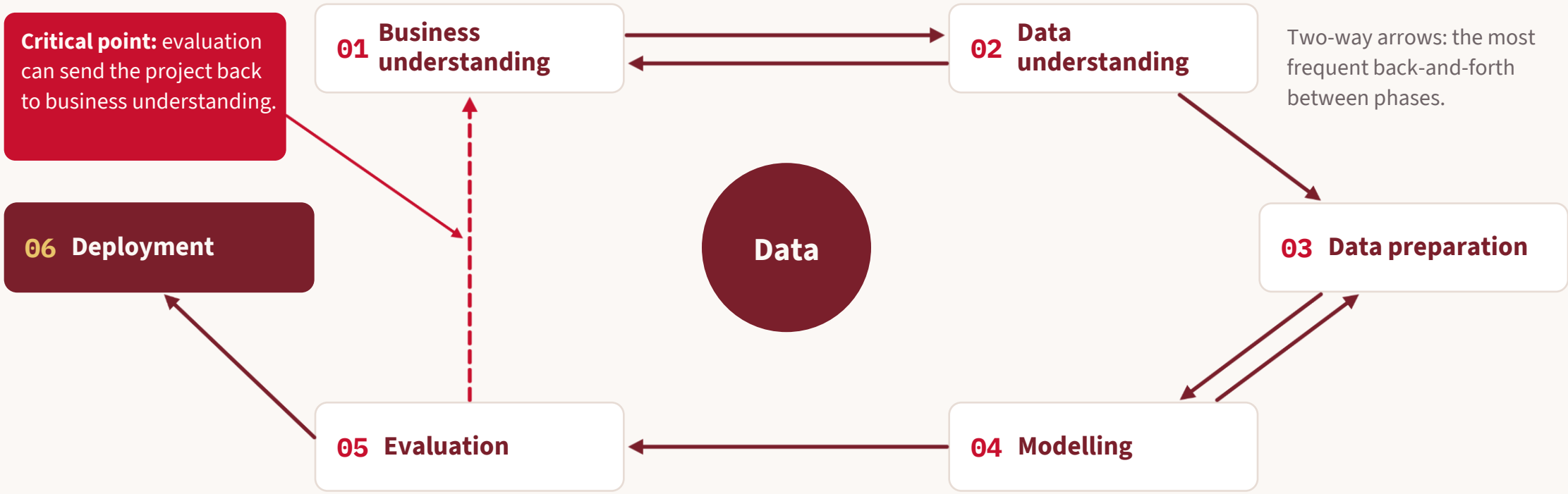
- Daimler-Benz (later DaimlerChrysler)
- ISL, later part of SPSS
- NCR
- OHRA, a Dutch insurer

In practice

Wirth & Hipp (2000), this week's required paper, describes the model and why it was built.

Objective 01

The Six Phases of CRISP-DM



- The phases surround the **data**: every phase works with it.
- The outer path is a **cycle**: deployment raises new business questions.

Phases 1–3: Generic Tasks and Outputs

CRISP-DM lists **generic tasks** for every phase, and the output each task should produce.

Phase	Generic tasks	Key outputs
1 Business understanding	Determine business objectives · assess situation · determine data mining goals · produce project plan	Business and data mining success criteria; project plan
2 Data understanding	Collect initial data · describe data · explore data · verify data quality	Data description and data quality reports
3 Data preparation	Select data · clean data · construct data · integrate data · format data	Final dataset and its description

In practice

These outputs are a checklist for the methodology section of your project report.

Critical point: two kinds of success criteria: one for the business, one for the model.

Phases 4–6: Generic Tasks and Outputs

Phase	Generic tasks	Key outputs
4 Modelling	Select modelling technique · generate test design · build model · assess model	Test design; models with their settings and assessment
5 Evaluation	Evaluate results · review process · determine next steps	Assessment against business criteria; decision to deploy, iterate or stop
6 Deployment	Plan deployment · plan monitoring and maintenance · produce final report · review project	Deployment and monitoring plans; final report; lessons learned

Common mistake: choosing the evaluation measure after seeing which one makes the model look best.

Note: the test design is decided **before** any model is built.

Key point: decide how models will be judged before building them, and finish every project by recording what was learned.

Objective 02

Mapping IBM's Ten Stages to CRISP-DM

CRISP-DM phase	IBM Foundational Methodology stages
1 Business understanding	1 Business understanding · 2 Analytic approach
2 Data understanding	3 Data requirements · 4 Data collection · 5 Data understanding
3 Data preparation	6 Data preparation
4 Modelling	7 Modelling
5 Evaluation	8 Evaluation
6 Deployment	9 Deployment · 10 Feedback

In practice

Your project may use either model, but must name it and follow its structure.

Why? IBM makes feedback a stage of its own; CRISP-DM plans monitoring inside deployment.

Key point: IBM has ten finer stages and an explicit feedback stage; CRISP-DM has six phases with defined outputs. The logic is the same

Objective 04

A process model is a checklist for thinking, not a timetable.

CRISP-DM and IBM's stages list the questions every project must answer. They do not require one long pass in a fixed order, and they do not replace judgement about what a project needs.

Rule of thumb: use the phases to check what is missing, then work in short loops through all of them.

Criticisms of CRISP-DM

Martínez-Plumed et al. (2021) review how well the model fits data science twenty years later:

- It was built for **goal-directed** data mining; exploratory work and data products fit poorly.
- It names few **roles**: who owns data engineering, deployment or monitoring?
- It is thin on **operations**: versioning, automated retraining and drift detection.
- It has no explicit checkpoint for **ethics**, privacy or fairness.



The authors propose to extend CRISP-DM with new trajectories, not to abandon it.

Summary of Martínez-Plumed et al., IEEE TKDE 33(8), 2021



Common mistake: concluding that CRISP-DM is obsolete. Most gaps are closed by adding practices around it, not by replacing it.

Other Process Models

KDD process (Fayyad et al., 1996)

Knowledge discovery in databases: selection, preprocessing, transformation, data mining, interpretation and evaluation. Academic; little on business or deployment.

SEMMA (SAS)

Sample, Explore, Modify, Model, Assess. A practical modelling sequence for SAS tools; assumes the business problem is already defined.

TDSP (Microsoft)

Team Data Science Process: business understanding, data acquisition and understanding, modelling, deployment, customer acceptance. Adds roles and templates.

MLOps

DevOps practices for machine learning: versioned data and models, automated tests and retraining, continuous deployment, monitoring.

Key point: KDD and SEMMA cover less than CRISP-DM; TDSP and MLOps extend it towards teams and operations.

Comparing Process Models

Model	Origin	Main strength	Main gap
KDD	Academic, 1996	Clear technical steps	Business context, deployment
SEMMA	SAS, vendor	Practical modelling sequence	Business problem; tied to one tool
CRISP-DM	Industry consortium, 2000	Business focus, defined outputs	Roles, operations, ethics
IBM	IBM, 2015	Fine data stages, explicit feedback	Fewer defined outputs
TDSP	Microsoft, about 2016	Team roles, templates, agile	Leans on Microsoft tools
MLOps	Industry practice	Reliable models in production	Not a full project method

In practice

A common default: CRISP-DM or IBM for the project, TDSP roles for the team, MLOps for anything deployed.

Read the gaps column. No single model covers everything, so teams combine them.

Adding an Ethics Checkpoint

None of the classic models includes ethics explicitly. Every DS 501 project adds two checkpoints:

At business understanding

Who could be **harmed** by the decision this project supports?

Is there a lawful basis for the personal data, and is the data limited to what is necessary?

Would the people in the data reasonably expect this use?

At evaluation

Does the model work equally well for different groups, such as age, sex or region?

Can a decision be **explained** to the person it affects?

What happens when the model is wrong, and who can correct it?

In practice

GDPR principles include purpose limitation and data minimisation (Article 5). Week 12 covers privacy and fairness in depth.

Objective 05

Workshop: Frame a Real Problem with CRISP-DM

CHOOSE ONE SCENARIO

A · Skopje air quality: warn residents a day before PM2.5 exceeds the EU daily limit.

B · IBU student retention: identify first-year students at risk of dropping out.

C · Telecom churn: decide which customers receive a retention offer this month.

GROUPS OF 3–4 · 15 MIN

1. Write the business objective and its success criterion.
2. Write the data mining goal and its success criterion.
3. List four data requirements and where to obtain them.
4. Name one ethical risk and how you would reduce it.

Check Your Understanding

Before building any model, a churn team decides to compare models with 5-fold cross-validation and to use recall as the measure. In CRISP-DM, which phase and task is this?

A Business understanding: determine data mining goals.

B Data preparation: construct data.

C Modelling: generate test design.

D Evaluation: review process.

Think for 30 seconds, then vote with your fingers: 1 = A, 2 = B, 3 = C, 4 = D.

Key Terms

CRISP-DM

Cross-Industry Standard Process for Data Mining: six phases, published in 2000.

Data mining goal

The business objective restated in technical terms.

KDD process

Fayyad et al.'s steps from data selection to interpretation (1996).

MLOps

Engineering practices for versioning, deploying and monitoring models.

Generic task

A CRISP-DM task that applies to every project, with a defined output.

Test design

The plan, made before modelling, for how models will be judged.

TDSP

Microsoft's Team Data Science Process, which adds roles and templates.

Ethics checkpoint

Questions on harm, lawfulness and fairness at two fixed phases.

Summary

In this lesson, you learned how to:

- **O1** Describe the origin and six phases of CRISP-DM
- **O2** Distinguish CRISP-DM from IBM's methodology
- **O3** Compare other process models with CRISP-DM
- **O4** Evaluate the main criticisms of CRISP-DM
- **O5** Apply CRISP-DM with an ethics checkpoint

Key takeaway: use a process model as a checklist, combine models to cover their gaps, and always add the ethics checkpoints.

Next

Practice 2 · CRISP-DM walkthrough on telecom churn

Practice 2: CRISP-DM Walkthrough

You take one small project through all six CRISP-DM phases, writing the question before you touch the data.

Goal

Apply every CRISP-DM phase, find and fix a real data-quality issue, and judge a model by a business cost.

Dataset

IBM Telco Customer Churn: **7,043** customers, **21** columns, and whether each one left last month.

Notebook

W02_Lab_CRISP_DM_Walkthrough.ipynb in Google Colab. No installation is needed.

Time

60–75 minutes in class; finish the written answers at home before the week 3 session.

Phase 1 first

Complete the business understanding template before loading any data.

Simple on purpose

A baseline and a logistic regression only. The focus is the process; algorithms come in weeks 9–10.

Exercises are marked ✏️. Every number you report needs a sentence that interprets it.

Phase 2: Find the Data Quality Issue

```python
# Python · Google Colab
raw = pd.read_csv(URL)
print(raw.shape)                          # (7043, 21)
bad = raw[pd.to_numeric(raw["TotalCharges"],
                        errors="coerce").isna()]
print(len(bad), "rows cannot be converted")
print(raw["Churn"].value_counts(normalize=True).round(3))
```

1–2 Load the data from its public URL and check its size.

← **3–4 · Critical:** TotalCharges is stored as **text**; find the rows that are not numbers.

5 Prints 11: all eleven have a tenure of 0 months.

6 Churn rate: 0.265 left, 0.735 stayed.

⚠ **Common mistake:** filling the 11 blanks with the mean. New customers have not been billed yet, so the right value is 0.

Phases 3–4: Split First, Then Model

```
# Python · scikit-learn
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, stratify=y, random_state=42)
baseline = DummyClassifier(
    strategy="most_frequent").fit(X_train, y_train)
logit = make_pipeline(StandardScaler(),
    LogisticRegression(max_iter=1000)).fit(X_train, y_train)
```

1–2 · Critical: split **before** fitting anything; stratify keeps 26.5% churners in both sets.

3–4 Baseline: always predicts “stays”. Test accuracy 0.735.

5–6 Scaling and logistic regression in one pipeline. Test accuracy 0.806.

</> In practice

Inside a pipeline, the scaler learns its means only from the training data, which prevents one kind of leakage.

Phase 5: Evaluate Against a Business Cost

```python
# Python · €40 offer, €300 lost customer (illustrative)
proba = logit.predict_proba(X_test)[:, 1]
def cost(t, retain=RETAIN_RATE):
    pred = (proba >= t).astype(int)
    tn, fp, fn, tp = confusion_matrix(y_test, pred).ravel()
    return int((tp + fp) * OFFER_COST
               + (fn + tp * (1 - retain)) * LOST_VALUE)
```

1 The probability of leaving for each test customer. ROC AUC 0.846.

2–4 For a threshold t, count caught, missed and wasted cases.

5–6 · Measure: offers cost €40 each; every churner not kept costs €300. Lowest cost: threshold **0.25**.

Key point: contacting nobody costs €140,100 on the test set; the best threshold costs €112,940.

What to Submit

Submit the completed notebook (.ipynb) on the course page. It must run from top to bottom without errors.

Part	What is marked	Points
Run	The notebook runs from top to bottom (Restart and run all)	2
Phase 1	Business template completed before the data, with both success criteria	2
A	The TotalCharges issue explained; churn rates interpreted as correlations	2
B–C	Leakage example; baseline compared; cost analysis with RETAIN_RATE = 0.3	2
Phase 6	Deployment, monitoring and ethics plan in about 150 words	2
	Total	10

⚠ **Common mistake:** reporting the 0.806 accuracy without the 0.735 baseline beside it.

← **Due:** [date], before the week 3 session.

WEEK 2 · PAPERS TO READ

Papers to Read

The paper that introduced CRISP-DM, an argument for data-driven decision making, and three papers on how the process has been described, criticised and extended.

Required

Required Reading for Week 3

Read the listed sections before the week 3 class. One question in Quiz 2 may refer to these papers.

CRISP-DM: Towards a Standard Process Model for Data Mining

Wirth & Hipp · Proc. 4th Int. Conf. on the Practical Applications of Knowledge Discovery and Data Mining (PADD), 2000

The source of the six phases. Read the whole paper; it is short.

Data Science and its Relationship to Big Data and Data-Driven Decision Making

Provost & Fawcett · Big Data 1(1), 2013

Why data science exists to improve decisions. Read the whole essay.

Bring a half-page note per required paper: the problem, the key idea, and its link to this week.

Optional

Optional Reading

From Data Mining to Knowledge Discovery in Databases
Fayyad, Piatetsky-Shapiro & Smyth · AI Magazine 17(3), 1996

The KDD process. Read the sections on the KDD steps.

CRISP-DM Twenty Years Later: From Data Mining Processes to Data Science Trajectories
Martínez-Plumed et al. · IEEE TKDE 33(8), 2021

Criticisms and the trajectories model. Read the introduction and conclusions.

Hidden Technical Debt in Machine Learning Systems
Sculley et al. · NeurIPS (NIPS 28), 2015

Why models in production are hard to maintain. Eight pages.

Optional papers are not examined. Choose one that relates to your project.

WEEK 2 · REQUIRED READING 1 · WIRTH & HIPP 2000

Required

A Standard Process for Data Mining Projects

Wirth, R., & Hipp, J. (2000). CRISP-DM: Towards a standard process model for data mining. In Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining (pp. 29–39).

THE PROBLEM

In 2000 there was no standard way to run a data mining project. Success depended on the people doing it, and good practice was hard to repeat. The authors argue for a standard process and test CRISP-DM in practice.

KEY IDEAS

01 Six phases, from business understanding to deployment; the order between them is not strict

02 Four levels: phases, generic tasks, specialised tasks, and process instances (records of real work)

03 The Reference Model says what to do; the User Guide gives tips on how to do it

Key term: Generic task — A task general enough for any data mining project, e.g. build model; a specialised task adapts it to one case

Required

CRISP-DM Works, but Plan for Iterations Explicitly

WHAT TO TAKE AWAY

1. Using the model to communicate helped more than expected and built trust with users and sponsors
2. Skipping planning and documentation to save time cost the expert team more time later
3. Data preparation took longer than expected; they suggest planning three iterations, each shorter

THIS WEEK

The source of the six phases and generic tasks in lesson 2-3. In Practice 2 you take Telco churn through the same phases and write the business template before loading data.

LIMITATION

An experience report from one company and one application. The authors admit that the claimed gains in speed and cost are hard to evaluate.

For your half-page note: The paper lists benefits of a standard process for customers, vendors and analysts. Which one matters most to you, and why?

Required

Data Science Exists to Improve Decisions

Provost, F., & Fawcett, T. (2013). Data science and its relationship to big data and data-driven decision making. Big Data, 1(1), 51–59. https://doi.org/10.1089/big.2013.1508

THE PROBLEM

By 2013, data science was a popular term, but people disagreed about what it meant. The authors fear it could become an empty buzzword, so they relate it to big data and to data-driven decision making.

KEY IDEAS

01 Data science is a set of principles for extracting knowledge from data, not a set of tools

02 Its main goal is better decisions: some are one-off discoveries, others repeat at massive scale

03 Big data technologies support data science, but much data processing is not data science

Key term: Data-analytic thinking — Viewing business problems from a data perspective, using the fundamental principles of data science

Required

Learn the Principles; the Tools Will Change

WHAT TO TAKE AWAY

1. Cites a study: one standard deviation more data-driven decisions went with 4–6% higher productivity
2. Lists principles: follow a staged process such as CRISP-DM, evaluate in context, use expected value
3. Warns about overfitting, and that causal conclusions need close attention to confounding factors

THIS WEEK

Its MegaTelCo churn example is the problem of Practice 2. Its expected-value principle is the idea behind the €40 offer and €300 lost customer cost analysis in phase 5.

LIMITATION

A short essay, not a study. Its evidence on firm performance comes from other authors' working papers; its list of principles is partial.

For your half-page note: The authors say results must be evaluated in the context where they will be used. How does Practice 2 do this for the churn model?

Optional

Data Mining Is One Step in a Longer Process

Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). From data mining to knowledge discovery in databases. AI Magazine, 17(3), 37–54. https://doi.org/10.1609/aimag.v17i3.1230

THE PROBLEM

In the mid-1990s, data grew faster than people could analyse it by hand, and terms such as data mining were used loosely. The authors define knowledge discovery in databases and describe its whole process.

KEY IDEAS

01 KDD is the whole process of finding useful knowledge; data mining is one step within it

02 Found patterns should be valid on new data, novel, potentially useful and understandable

03 Mining algorithms combine three parts: model representation, model evaluation and search

Key term: Data dredging — Blindly applying mining methods until something appears; it easily finds meaningless or invalid patterns

Optional

Most of the Work Lies Outside the Algorithm

WHAT TO TAKE AWAY

1. Describes nine steps, from understanding the domain and goal to acting on the knowledge found
2. Most research focused on the mining step, but the other steps matter as much or more in practice
3. Much application effort goes into formulating the problem well, not into tuning the algorithm

THIS WEEK

The KDD process in lesson 2-3's comparison of process models. Compare its steps with CRISP-DM's six phases and the ten IBM stages from lessons 2-1 and 2-2.

LIMITATION

A 1996 overview for an AI audience. Its survey of methods is now dated, and it gives no guidance on managing a project or a team.

To think about: The paper's figure starts at data selection, but its list of steps starts with the goal. Which view does a methodology need?

Optional

From One Process to Many Data Science Trajectories

Martínez-Plumed, F., et al. (2021). CRISP-DM twenty years later: From data mining processes to data science trajectories. IEEE Trans. Knowl. Data Eng., 33(8), 3048–3061. https://doi.org/10.1109/TKDE.2019.2962680

THE PROBLEM

CRISP-DM is about twenty years old and still the most used process model, but data science has become more varied and exploratory. The authors ask whether, and for which projects, CRISP-DM still fits.

KEY IDEAS

01 Goal-directed projects still fit CRISP-DM; exploratory projects need a more flexible model

02 The DST map adds six exploration and four data management activities to CRISP-DM's six phases

03 A trajectory is the path one project takes through these activities, in no fixed order

Key term: Trajectory — The sequence of activities one project follows on the DST map; it may skip, repeat or run steps in parallel

Optional

CRISP-DM Still Fits, but Only One Kind of Project

WHAT TO TAKE AWAY

1. Seven real examples, such as a parking app and a data repository, use only parts of CRISP-DM
2. Of 51 NIST big data use cases, 23 involve data management activities only
3. Exploration needs senior data scientists and is hard to cost; data management needs engineers

THIS WEEK

Background to the criticisms of CRISP-DM in lesson 2-3. Its three project types can help you describe your own idea for project step 2.

LIMITATION

The trajectories are the authors' reconstructions of possible paths, not project records. They say DST does not yet address ethics fully.

To think about: Is your project idea goal-directed, exploratory or mainly data management? Which activities outside CRISP-DM might it need?

Optional

ML Systems Are Cheap to Build, Costly to Maintain

Sculley, D., et al. (2015). Hidden technical debt in machine learning systems. Advances in Neural Information Processing Systems, 28. https://proceedings.neurips.cc/paper/2015/hash/86df7dcfd896fcaf2674f757a2463eba-Abstract.html

THE PROBLEM

Engineers at Google found that machine learning systems are fast and cheap to build and deploy, but difficult and expensive to maintain over time. The authors explain why, using the idea of technical debt.

KEY IDEAS

01 Models mix all their inputs, so changing anything changes everything (the CACE principle)

02 Data dependencies build up debt like code dependencies, but they are harder to detect

03 Model code is a small part of a real ML system; the surrounding infrastructure is vast

Key term: Technical debt — Future maintenance cost created by moving quickly now; like financial debt, it grows if it is not paid down

Optional

Monitor the Live System, Not Only the Model

WHAT TO TAKE AWAY

1. Feedback loops form when a model's predictions influence the data it is later trained on
2. A manually set decision threshold can become invalid when the model is retrained on new data
3. Suggests monitoring prediction bias, setting action limits, and checking upstream data producers

THIS WEEK

Explains why lesson 2-3 adds MLOps to CRISP-DM. Useful for the monitoring plan in Practice 2, phase 6, and for the decision threshold you choose in phase 5.

LIMITATION

Based on the authors' experience, not on measured data. It names many risks but, as the authors admit, offers no metric for technical debt.

To think about: Your churn threshold rests on cost assumptions. What could make it wrong a year after deployment, and how would you notice?

Discussion

Reading Questions for Discussion

- Wirth and Hipp wanted a process that was industry-neutral and tool-neutral. Why would two tool vendors in the consortium support that?
- Provost and Fawcett define data science through decisions. Which IBM stage puts that idea into practice?
- Which criticism by Martínez-Plumed et al. would matter most in your own organisation, and why?
- Is the KDD process a methodology in the sense of lesson 2-1? Why or why not?

Prepare for class

- Read the assigned papers
- Write a half-page note per paper
- Bring one answer to each question
- Discuss in groups of three

Support each answer with one specific passage from the paper, quoted or paraphrased.

DS 501 · WEEK 2

This Week's Work

Practice 2 notebook

Finish W02_Lab_CRISP_DM_Walkthrough.ipynb, check that it runs from top to bottom, and submit the .ipynb file.

Due: [date]

Quiz 2

15 multiple-choice questions on lessons 2-1 to 2-3, plus one short applied question on framing a problem.

Due: [date]

Project step 2

On one page, frame your project idea: business understanding checklist, analytic approach, data requirements, one ethical risk.

Due: [date]

Next week: Data Sources, Formats and Acquisition: where data comes from, how it is stored and described, and how to collect it with Python from files, APIs and web pages.